RAJALAKSHMI ENGINEERING COLLEGE

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M2

Started on	Thursday, 24 September 2026, 11:03 AM
State	Finished
Completed on	Thursday, 24 September 2026, 11:13 AM
Time taken	10 mins 6 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

int main() {
    char s1[1001], s2[1001];

    scanf("%s", s1);
    scanf("%s", s2);

    int n = strlen(s1);
    int m = strlen(s2);

    int **dp = (int **)malloc((n + 1) * sizeof(int *));

    for (int i = 0; i <= n; i++) {
        dp[i] = (int *)calloc(m + 1, sizeof(int));
    }

    // Build the LCS table
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) {
            if (s1[i - 1] == s2[j - 1]) {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            } else {
                dp[i][j] = (dp[i - 1][j] > dp[i][j - 1])
                           ? dp[i - 1][j]
                           : dp[i][j - 1];
            }
        }
    }

    printf("%d", dp[n][m]);

    for (int i = 0; i <= n; i++) {
```

```
36          free(dp[i]);
37      }
38      free(dp);
39
40      return 0;
41  }
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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